INCH-POUND

A-A-59707
10 December 2001

COMMERCIAL ITEM DESCRIPTION

DETECTOR/MONITOR, HYDRAZINE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE.

This commercial item description covers a continuous detector/monitor of toxic amounts of hydrazine vapors in open and enclosed areas. The detector is intended for bench top use to monitor the level of hydrazine present in working environments when servicing, maintaining, and overhauling aircraft components and assemblies in an immediate work area where exposure to H-70 (70% hydrazine, 30% water) hydrazine vapors exist.

2. SALIENT CHARACTERISTICS.

2.1 Design.

The detector shall be designed for bench top use and mounting. The detector maximum weight and volume should be approximately 40 pounds and 4000 cubic inches. It shall include all components, parts, and features required to meet these salient performance requirements.

2.2 Measurement system.

The detector measurement shall not incorporate any components that are controlled by any regulatory rules or commissions.

2.3 Measurement capability.

The detector shall be capable of continuously detecting and measuring hydrazine vapors ranging from 0 to 2.0 parts per million (ppm) in 0.01 ppm increments.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data, that may improve this document should be sent to: ASC/ENOI, 2530 Loop Road West, Bldg 560, Wright-Patterson AFB OH 45433-7101.

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2.4 Measurement distance.

The detector shall be capable of detecting the hydrazine vapor levels stated in 2.3 from distances of up to 6 inches.

2.4.1 Time to alarm.

The detector shall be capable of alarming to hydrazine detection at the following concentrations and times or to higher commercial standards.

TABLE I. Hydrazine detection alarm.

Hydrazine concentration	Time to alarm
2.0 ppm	15 seconds
1.0 ppm	30 seconds
0.5 ppm	60 seconds
0.1 ppm	270 seconds
0.01 ppm	900 seconds

2.4.2 Alarm points.

The detector shall be capable of providing a minimum of 2 alarm levels. The alarm settings shall be at the threshold limit value (TLV) and lower to meet base environmental warning and evacuation requirements.

2.5 Reliability and maintainability.

The detector shall have a meantime between failure and repair rating typical to industry grade gas detectors. The manufacturer shall provide commercial manuals required for operations, maintenance, and troubleshooting.

2.6 Display.

The detector shall provide an illuminated digital display readable at a minimum distance of 5 feet and in increments of 0.01 ppms.

2.7 Accuracy.

The detector shall have an accuracy of at least ±10% with repeatability equal to or greater than ±5% at TLV.

2.8 Alarm.

The detector alarm shall be audible to at least 80 decibels at a distance of 5 feet.

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2.9 Outputs.

As an optional requirement, the detector should provide an alarm signal to an RS232 and RS422 connection.

2.10 Operating temperatures.

The detector shall operate in a temperature range of 32-104 °F.

2.11 Relative humidity.

The detector shall operate to the specified performance and accuracy while exposed to environments with a 30 to 70% relative humidity.

2.12 Electrical.

The detector shall operate on 120 VAC, ±10%, and 220VAC, 50-60 Hertz, with a maximum power drain of 500 watts.

2.13 Safety.

The detector shall be capable of operating safely in a class I, division 2 location and adhere to National Fire Protection Agency (NFPA) requirements for commercial and industry use.

3. REGULATORY REQUIREMENTS

3.1 Recoverable materials.

Government policy on the use of recovered materials considers cost, availability of competition, and performance. In accordance with section 23.403 of the Federal Acquisition Regulation (FAR), the Federal Government's policy is to acquire competitively, in a cost-effective manner, products that meet reasonable performance requirements and that are composed of the highest percentage of recoverable materials practicable.

4. PRODUCT CONFORMANCE

4.1 Product conformance.

The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Metric products.

Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using the inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met. If the manufacturer elects to use metric dimensions as the basis upon which to derive or base a mechanical feature, a request shall be made to through the contracting office to the cognizant engineer to determine if the product is acceptable. The engineering activity has the option of accepting or rejecting the product.

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4.3 Warranty.

Unless otherwise specified in the contract or purchase order, the manufacturer's standard commercial warranty shall apply.

4.4 Market acceptability.

The company producing the item must have produced the item or a product similar to the item described in the past ten years or be able to provide independent data substantiating the product claim to meeting this commercial item description. The government reserves the right to require proof of productions date prior to first delivery.

5. PACKAGING.

Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Source of documents.

6.1.1 Federal Acquisition Regulation (FAR).

FAR 23.403 may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

6.1.2 FED-STD-376.

FED-STD-376, Preferred Metric Units for General Use by the Federal Government, may be obtained from the Defense Automation and Production Service (DAPS), Bldg 4D, 400 Robbins Ave., Philadelphia PA 19111-5094.

6.2 Ordering data.

The contract or order should specify the following:

a. CID title, number, and date.

b. Product conformance provisions.

c. Packaging requirements.

6.3 Key words.

alarm points

gas detectors

parts per million (ppm)

threshold limit value

vapor levels

vapors

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CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army - EA
Navy - SH
Air Force - 11
DLA -GS1

Preparing Activity:
AF - 11

Review Activity:
Navy - CG
Air Force - 99

Project No. 6665-0721

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